Spontaneous RNA Formation and the First Life on Earth

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What am I talking about

- DNA is made by copying other DNA. RNA is half of DNA
- Bootstrap paradox: where did the first DNA come from?
- How can we replicate these conditions?

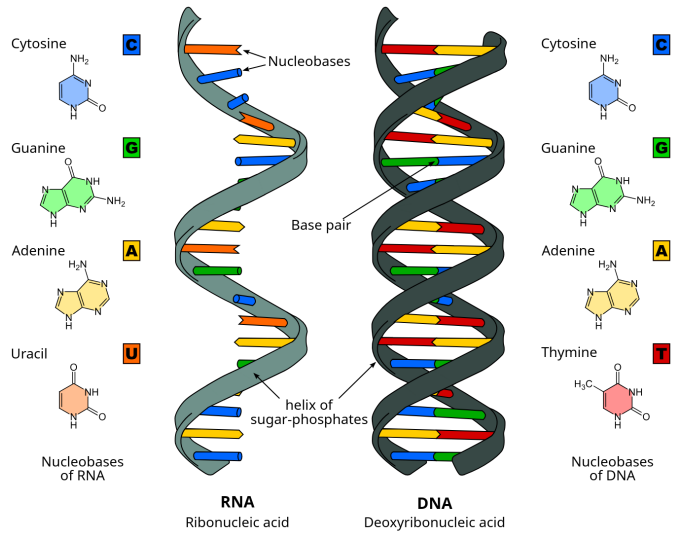


Figure: RNA/DNA structures.

Thermodynamic/Energetic Barriers

- Spontaneous organization of *nucleotides* is limited energetically and physically
- Want to bond phosphate-to-phosphate, but they may not be pointed at each other
- Also *endergonic*, releases water at every bond formation
- Also anti-entropic ($\Delta S < 0$), combining molecules increases order

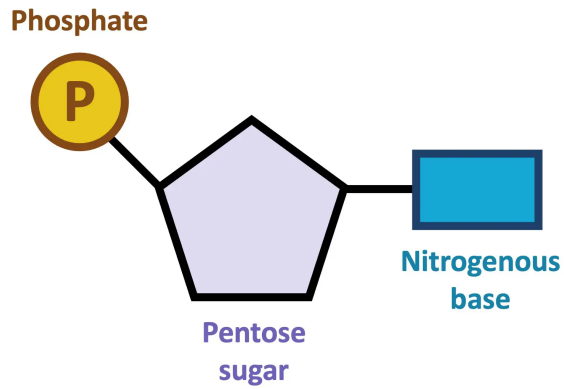


Figure: A nucleotide.

Encouragement

- Maybe heat? Temperatures required would destroy the nucleotides
- Reagents? Unlikely to be responsible for proto-RNA; highly complex chemically
- To thermodynamics!

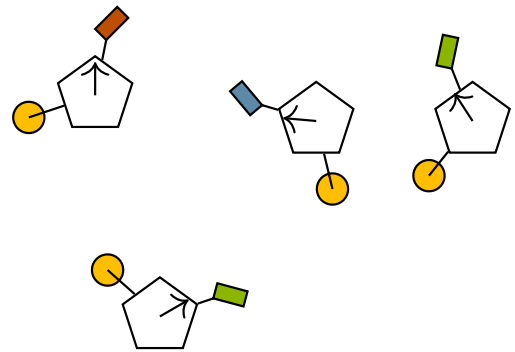


Figure: A bunch of nucleotides whizzing around.

Constraint

- We pre-organize the molecules
- Force them onto the surface of a sphere
- Decreases the Gibbs free energy

$$G = U + PV - TS$$

by decreasing degrees of freedom

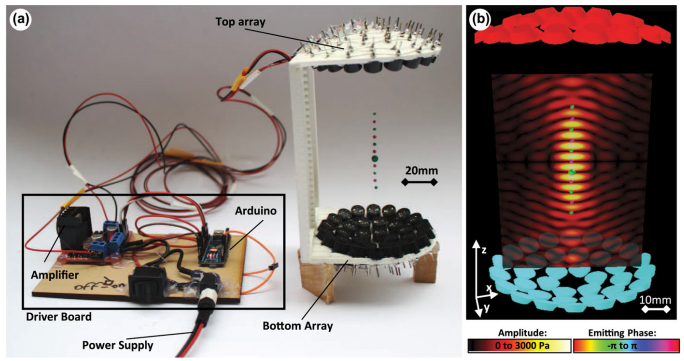


Figure: An ultrasonic levitator.

Constraint

- Levitating spheres of nucleotide containing water
- Creates a spontaneous reaction, allowing nucleotides to bond
- Entirely possible in early-Earth conditions; raindrops, wave spray

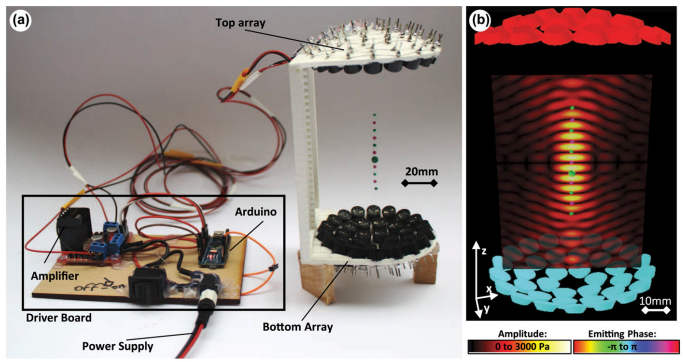


Figure: An ultrasonic levitator.

Summary

- Nucleotide bonding into DNA/RNA energetically unfavourable, physically unlikely
- Constrain nucleotides to a 2D surface
- Enables spontaneous bonding and DNA/RNA formation
- Ultrasonic levitators are much less complex than other solutions



Figure: Non-levitating wet-dry cycle device.